$\sin \omega t \Rightarrow \frac{\omega}{S^2+\omega^2}$

$\cos \omega t \Rightarrow \frac{S}{S^2+\omega^2}$

$e^{-at} \sin \omega t \Rightarrow \frac{\omega}{(S+a)^2+\omega^2}$

$e^{-at} \cos \omega t \Rightarrow \frac{(S+a)}{(S+a)^2+\omega^2}$

Laplace Transforms

$f(t)$	$F(S)$
$\delta(t)$ (unit impulse function)	1
$u(t)$ (unit step function)	$\frac{1}{S}$
t (unit ramp function)	$\frac{1}{S^2}$
t^n	$\frac{n!}{S^{n+1}}$
e^{-at} عكس الإشارة	$\frac{1}{s+a}$
$t^n e^{-at}$	$\frac{n!}{(S+a)^{n+1}}$
$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at} \cos \omega t$	$\frac{(S+a)}{(s+a)^2+\omega^2}$
$\sinh \omega t$	$\frac{\omega}{s^2-\omega^2}$
$\cosh \omega t$	$\frac{S}{s^2-\omega^2}$
$\frac{d(f)}{d(t)} = f^1(t)$	$SF(s) - f(0)$
$\frac{d^2(f)}{d(t)^2} = f^2(t)$	$S^2F(s) - Sf(0) - f^{\cdot}(0)$
$\frac{d^3(f)}{d(t)^3} = f^3(t)$	$S^3F(s) - S^2f(0) - Sf^{\cdot}(0) - f^{\cdot\cdot}(0)$
$\frac{d^n(f)}{d(t)^n} = f^n(t)$	$S^nF(s) - S^{n-1}f(0) - S^{n-2}f^1(0) - \cdots - f^{n-1}(0)$
$\int f(t)dt$	$\frac{1}{S}F(s)$

$S = \alpha + j\omega$

$t^3 = \frac{3!}{S^{3+1}} = \frac{6}{S^4}$

$t^2 e^{3t} \rightarrow \frac{2!}{(S-1)^3}$

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Control Engineering Fundamentals

Chapter 2

Laplace Transform & Transfer Function of The Electrical System

2.1 Definition of Laplace Transform

The Laplace transform is defined as below :

Let f(t) be a real function of a real variable t defined for t > 0, then

$$F(s) = L[f(t)] = \int_0^\infty f(t).e^{-st}\,dt$$

where F(s) is called Laplace transform of f(t). And the variable 's' which appears in F(s) is frequency dependent complex variable.

The time function f(t) is obtained back from the Laplace transform by a process called Inverse Laplace transform and denoted as L^{-1}.

$$L^{-1}[F(s)] = L^{-1}[L(f(t))] = f(t)$$

Example 2.1 : *Find the Laplace transform of e^{-at} and 1 for $t \geq 0$.*

Solution : i) $f(t) = e^{-at}$

$$F(s) = L[f(t)] = \int_0^\infty f(t)\,e^{-st}\,dt = \int_0^\infty e^{-at}\cdot e^{-st}\,dt$$

$$= \int_0^\infty e^{-(s+a)t}\,dt = \left[-\frac{1}{(s+a)}\cdot e^{-(s+a)t}\right]_0^\infty$$

$$= 0 - \left(\frac{-1}{s+a}\right) = \frac{1}{s+a}$$

$$\therefore \quad L[e^{-at}] = \frac{1}{s+a} \quad \text{and} \quad L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$$

(ii) $f(t) = 1$

$$\therefore \quad F(s) = \int_0^\infty f(t)\,e^{-st}\,dt = \int_0^\infty 1\,e^{-st}\,dt = \left[\frac{e^{-st}}{-s}\right]_0^\infty = \frac{1}{s}$$

$$\therefore \quad L\{1\} = \frac{1}{s} \quad \text{and} \quad L^{-1}\left\{\frac{1}{s}\right\} = 1$$

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2.2 Solution of Differential Equation Using Laplace Transform

In general, the procedure of solving differential equations by ***Laplace Transform*** consists of the following steps:

1- Take ***Laplace Transform*** of each term in the differential equation.
This step eliminates the time variable & its derivatives from the original equation ***Time Domain (t)*** & results in an algebraic equation in the ***Frequency Domain (S).***

2- Solve the resulting algebraic equation in the ***Frequency Domain (S).***

3- Obtain the **inverse** of ***Laplace Transform.*** This final step will give you the solution of the differential equation.

Example (2.2): Consider the following equation:

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 4y(t) = 2e^{-2t}$$

Find the solution corresponding to zero initial conduction:

$$y(0) = y'(0) = 0$$

Solution:

Tacking Laplace Transform for both sides will give us:

$$S^2Y(S) - Sy(0) - y'(0) + 5SY(S) - 5y(0) + 4Y(S) = \frac{2}{S+2}$$

Due to the zero initial conduction

$$S^2Y(s) + 5SY(s) + 4Y(s) = \frac{2}{S+2}$$

$$Y(s)\,(S^2 + 5S + 4) = \frac{2}{S+2}$$

$$Y(s) = \frac{2}{(S+2)(S^2+5S+4)}$$

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$$Y(s) = \frac{2}{(S+2)(s+1)(s+4)}$$

By using partial fraction expansion for Y(s) we'll have:

$$Y(S) = \frac{C_1}{(S+2)} + \frac{C_2}{(S+1)} + \frac{C_3}{(S+4)}$$

$$C_1 = \lim_{s\to -2} \frac{2}{(S+1)(S+4)} = -1$$

نعوض بدل S بـ (-2)

$$C_2 = \lim_{s\to -1} \frac{2}{(S+2)(S+4)} = \frac{2}{3}$$

$$C_3 = \lim_{s\to -4} \frac{2}{(S+2)(S+1)} = \frac{1}{3}$$

$$So\ Y(S) = \frac{C_1}{(S+2)} + \frac{C_2}{(S+1)} + \frac{C_3}{(S+4)}$$

$$Y(S) = -\frac{1}{(S+2)} + \frac{2}{3}\frac{1}{(S+1)} + \frac{1}{3}\frac{1}{(S+4)}$$

SO, take the inverse laplace transforms the solution will be:

$$y(t) = -e^{-2t} + \frac{2}{3}e^{-t} + \frac{1}{3}e^{-4t}$$

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Use of Laplace Transform in Control System

Example (2.3): Obtain the expression for y(t) which is satisfying the differential equation the inverse Laplace transform of,

$$\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 8y(t) = 16e^{-t} \quad \textbf{, Neglect initial conduction.}$$

Solution: *Taking Laplace transform of both sides of the given differential equation and neglecting initial conduction in Laplace transform, So we get,*

$$S^2Y(s) + 6SY(s) + 8Y(s) = \frac{16}{S+1}$$

$$Y(s)\,[\,S^2 + 6S + 8] = \frac{16}{S+1}$$

$$Y(s) = \frac{16}{(S+1)(S^2+6S+8)}$$

$$Y(s) = \frac{16}{(S+1)(s+2)(s+4)}$$

$$Y(S) = \frac{A}{(S+1)} + \frac{B}{(S+2)} + \frac{C}{(S+4)}$$

$$A = \lim_{s\to-1} \frac{16}{(S+2)(S+4)} = \frac{16}{(1)(3)} = \frac{16}{3} = 5.333$$

$$B = \lim_{s\to-2} \frac{16}{(S+1)(S+4)} = \frac{16}{(-1)(2)} = \frac{16}{-2} = -8$$

$$C = \lim_{s\to-4} \frac{16}{(S+1)(S+2)} = \frac{16}{(-3)(-2)} = \frac{16}{6} = 2.666$$

So $$Y(S) = \frac{A}{(S+1)} - \frac{B}{(S+2)} + \frac{C}{(S+4)}$$

$$Y(S) = \frac{5.33}{(S+1)} - \frac{8}{(S+2)} + \frac{2.66}{(S+4)}$$

$$Y(S) = 5.33\frac{1}{(S+1)} - 8\frac{1}{(S+2)} + 2.66\frac{1}{(S+4)}$$

SO, take the inverse LaPlace transforms the solution will be:

$$y(t) = 5.33e^{-t} - 8e^{-2t} + 2.66e^{-4t}$$

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Example (2.4): Find the inverse Laplace transform of,

$$F(s) = \frac{2S+5}{S^2+5S+6}$$

Solution:

Factorizing denominator,

$$F(s) = \frac{2S+5}{S^2+5S+6}$$

$$F(s) = \frac{2S+5}{(S+2)(S+3)} = \frac{A}{S+2} + \frac{B}{S+3} \quad \ldots Partial\ fractions$$

$$A = \lim_{s\to -2} \frac{2S+5}{(S+3)} = \frac{2(-2)+5}{(-2+3)} = \frac{1}{1} = 1$$

$$B = \lim_{s\to -3} \frac{2S+5}{(S+2)} = \frac{2(-3)+5}{(-3+2)} = \frac{-1}{-1} = 1$$

So $Y(S) = \frac{A}{(S+2)} + \frac{B}{(S+3)}$

$$Y(S) = \frac{1}{(S+2)} + \frac{1}{(S+3)}$$

SO, take the inverse laplace transforms the solution will be:

$$y(t) = e^{-2t} + e^{-3t}$$

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2.3 The Transfer Function

In these lectures we'll formulate the system representation by establish a definition of function that algebraically relates the system's output to its input.

This function allows us to algebraically combine the mathematical representations of subsystems to make a total system representation.

2.3.1 The definition of Transfer Function

It is the ratio of the output variable to the input variable, with all initial conditions assumed to be zero.

It is a property of the system elements only and it doesn't depend on the excitation (the input). As we'll see later, the transfer function characterizes the system's performance completely.

The transfer function can be represented as a block diagram with input & output, with the system's transfer function inside the block, as shown in Figure (3.1)

Transfer Function = Gain = $g(t)$; in time domain

= $G(S)$; in frequency domain

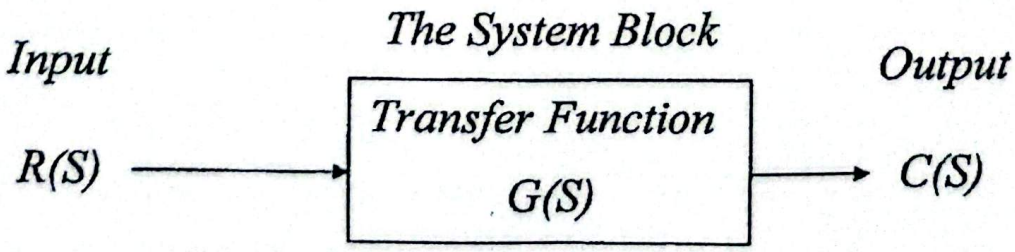


Figure (3.1)

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2.3.2 The Transfer Function of The Electrical System

Circuits with elements which store energy (inductors and capacitors) are dynamic systems (as opposed to static systems which have only resistors). The transfer function for a circuit can be obtained by substituting the Laplace value of the element's impedance:

Table (3.1) Summary of elements involved in linear electrical systems

Element	Voltage	Current
Resistor R (Ω: Ohm)	$v(t) = Ri(t)$	$i(t) = \frac{1}{R}v(t)$
Inductor L (H:Henry)	$v(t) = L\frac{di(t)}{dt}$	$i(t) = \frac{1}{L}\int_0^t v(\tau)d(\tau)$
Capacitor C (F:Farad)	$v(t) = \frac{1}{C}\int_0^t i(\tau)d(\tau)$	$i(t) = C\frac{dv(t)}{dt}$

The following set of symbols and units are used: $v(t) = V(Volts), i(t) = (Amps), q(t) = Q(Coulombs), C = F\,(Farads), R = \Omega\,(Ohms), L = H\,(Henries)$.

Table (3.2) Laplace value of the electric element

Element	Laplace value of Resistance	I=V/R Relationship	V=IR Relationship
Resistor	R	$I = \frac{V}{R}$	$V = IR$
Inductor	LS	$I = \frac{V}{LS}$	$V = I.LS$
Capacitor	$\frac{1}{CS}$	$I = V.CS$	$V = \frac{I}{CS}$

Current loops, voltage nodes, or other circuit analysis techniques can then be used to find the transfer function.

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2.4 Electrical and Electronic Systems

Basic laws governing electrical circuits are Kirchhoff's current law and voltage law.

Kirchhoff's current law (node law) states that the algebraic sum of all currents entering and leaving a node is zero. This law can also be stated as follows:

(The sum of currents entering a node is equal to the sum of currents leaving the same node).

Kirchhoff's voltage law (loop law) states that at any given instant the algebraic sum of the voltages around any loop in an electrical circuit is zero. This law can also be stated as follows:

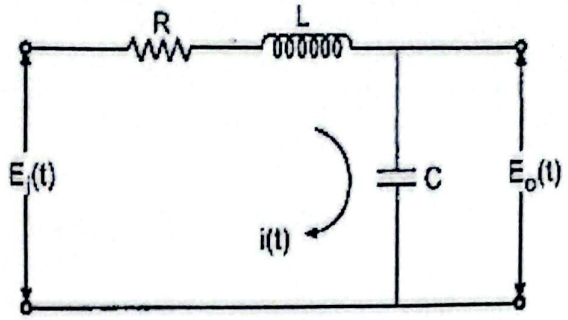
(The sum of the voltage drops is equal to the sum of the voltage rises around a loop.)

A mathematical model of an electrical circuit can be obtained by applying one or both of Kirchhoff's laws to it. This section first deal with simple electrical circuits and then treats mathematical modeling of systems.

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Examples/ Electrical System T.F

Example (2.5): *Find the T.F for the circuit shown below.*



Solution

$$E_i = iR + L\frac{di(t)}{dt} + \frac{1}{C}\int i\,dt \quad ---(1)$$

$$E_o = \frac{1}{C}\int i\,dt \quad ----(2)$$

Rewrite eq (1) in Laplace form

$$E_i = IR + LIS + \frac{I}{CS}$$

$$E_i = \left(R + LS + \frac{1}{CS}\right) \times I \quad ---(3)$$

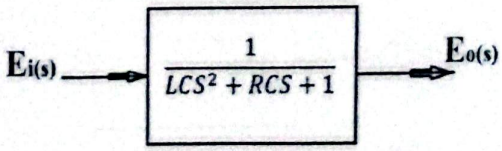
Rewrite eq (2) in Laplace form

$$E_o = \frac{I}{CS} \quad ---(4)$$

$$T.F = \frac{E_o}{E_i}$$

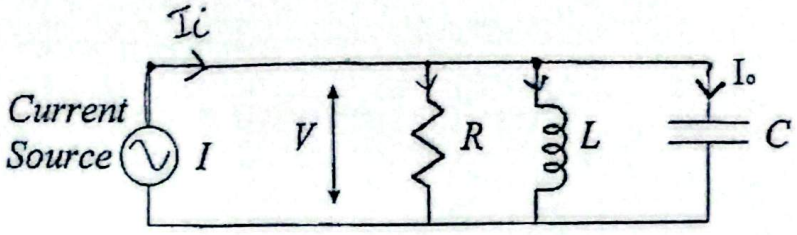
$$T.F = \frac{E_o}{E_i} = \frac{\frac{I}{CS}}{\left(R+LS+\frac{1}{CS}\right)\times I} = \frac{\frac{I}{CS}}{\frac{RCS+LCS^2+1}{CS}} = \frac{1}{CS}\,\frac{CS}{RCS+LCS^2+1}$$

$$\therefore T.F = \frac{E_o}{E_i} = \frac{1}{LCS^2 + RCS + 1}$$



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Example (2.6): *Find the T.F for the circuit shown below with the current source I (t).*



RLC Parallel Circuit

Solution:

$$i = \frac{v}{R} + \frac{1}{L}\int_0^t v\ dt + C\frac{dv(t)}{dt} \quad ---(1)$$

$$i_o = C\frac{dv(t)}{dt} \quad ----(2)$$

Rewrite eq (1) in Laplace form

$$I = \frac{V}{R} + \frac{V}{LS} + CS\,V$$

$$I = \left(\frac{1}{R} + \frac{1}{LS} + CS\right) \times V = I_i \quad ---(3)$$

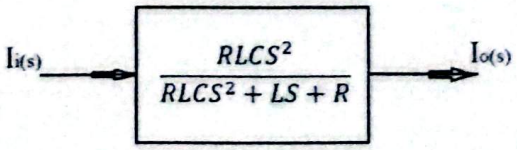
Rewrite eq (2) in Laplace form

$$I_o = CS \times V \quad ---(4)$$

$$T.F = \frac{I_o}{I_i}$$

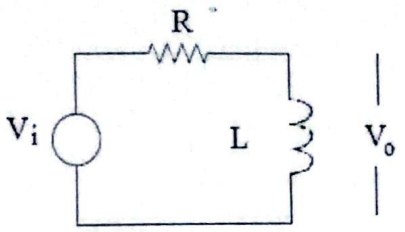
$$T.F = \frac{I_o}{I_i} = \frac{CS\,V}{\left(\frac{1}{R}+\frac{1}{LS}+CS\right)\times V} = \frac{CS}{\frac{LS+R+RLSCS}{RLS}} = \frac{RLSCS}{LS+R+RLCS^2}$$

$$\therefore\ T.F = \frac{I_o}{I_i} = \frac{RLCS^2}{RLCS^2 + LS + R}$$



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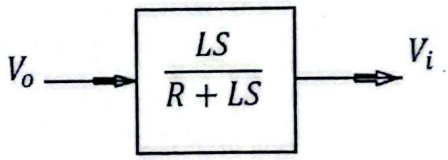
Example (2.7): Find the Transfer Function (V_o/V_i) for the systems shown below:



Solution

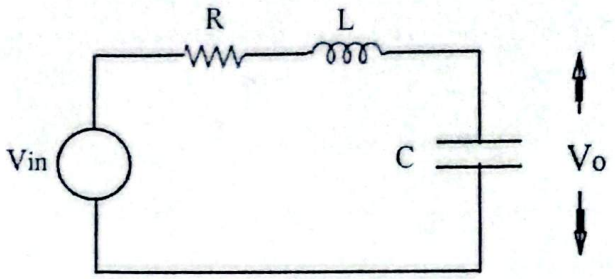
$$V_o = V_i \frac{LS}{R+LS}$$ Voltage divider rule

$$TF = \frac{V_o}{V_i} = \frac{LS}{R+LS}$$

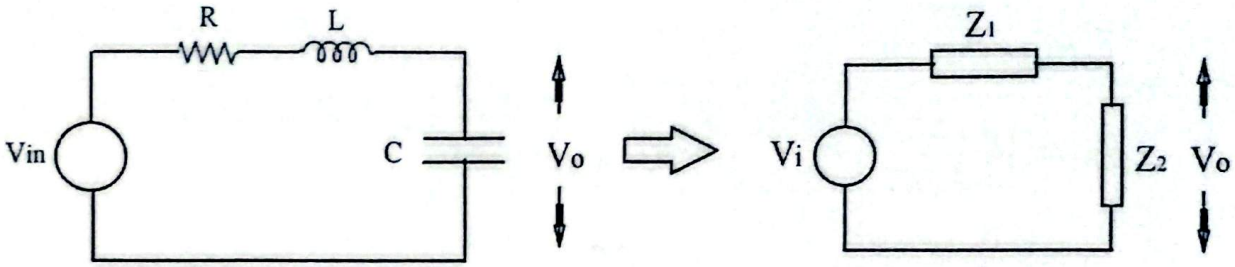


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Example (2.8): Find the Transfer Function (V_o/V_i) for the systems shown below.



Solution



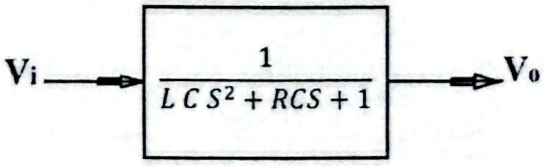
$$Z_1 = LS + R$$

$$Z_2 = \frac{1}{CS}$$

$$V_o = V_i \frac{Z_2}{Z_1+Z_2}$$ Voltage divider rule

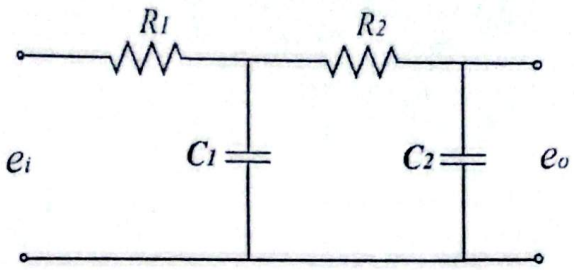
$$TF = \frac{V_o}{V_i} = \frac{Z_2}{Z_1+Z_2} = \frac{\frac{1}{CS}}{LS+R+\frac{1}{CS}} = \frac{\frac{1}{CS}}{\frac{LCS^2+RCS+1}{CS}}$$

$$TF = \frac{V_o}{V_i} = \frac{1}{LCS^2+RCS+1}$$



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Example (2.9) : *find the T.F.* Transfer Function (V_o/V_i) *for the given network:*

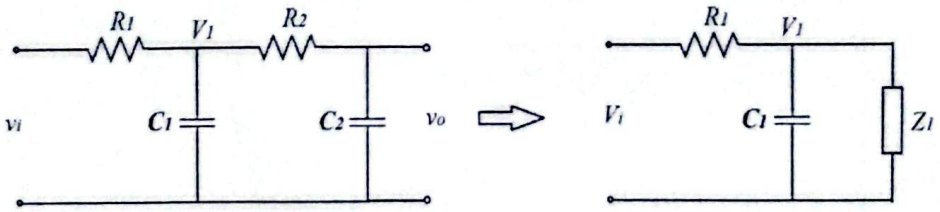


Solution:

$$V_o = V_1 \frac{\frac{1}{C_2 S}}{R_2 + \frac{1}{C_2 S}}$$ Voltage divider rule

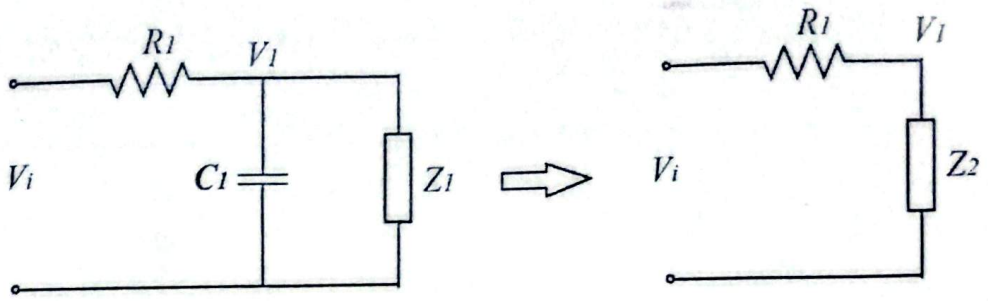
$$V_o = V_1 \frac{\frac{1}{C_2 S}}{\frac{R_2 C_2 S + 1}{C_2 S}} = \frac{1}{R_2 C_2 S + 1}$$

$$V_o = V_1 \frac{1}{R_2 C_2 S + 1} \quad \ldots\ldots (1)$$



$$Z_1 = R_2 + \frac{1}{C_2 S} = \frac{R_2 C_2 S + 1}{C_2 S}$$

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$$Z_2 = \frac{1}{C_1 S} // Z_1 = \frac{\frac{1}{C_1 S}\left(\frac{R_2 C_2 S+1}{C_2 S}\right)}{\frac{1}{C_1 S}+\frac{R_2 C_2 S+1}{C_2 S}} = \frac{\frac{R_2 C_2 S+1}{C_1 S \ C_2 S}}{\frac{C_2 S+R_2 C_2 S\, C_1 S+C_1 S}{C_1 S\, C_2 S}}$$

$$Z_2 = \frac{R_2 C_2 S + 1}{C_2 S + R_2\, C_1\, C_2\, S^2 + C_1\, S}$$

$$V_1 = V_i \frac{Z_2}{R_1 + Z_2} \quad \ldots\ldots (2)$$

Substitute equation (2) in (1):

$$V_o = V_1 \frac{1}{R_2\, C_2 S + 1} \quad \ldots\ldots (1)$$

$$V_o = V_i \ \frac{Z_2}{R_1 + Z_2} \ \ \frac{1}{R_2\, C_2 S + 1}$$

$$T.F. = \frac{V_o}{V_i} = \ \frac{Z_2}{R_1 + Z_2} \ \ \frac{1}{R_2\, C_2 S + 1}$$

$$\frac{V_o}{V_i} = \ \frac{\frac{R_2 C_2 S + 1}{C_2 S + R_2\, C_1\, C_2\, S^2 + C_1\, S}}{R_1 + \frac{R_2 C_2 S + 1}{C_2 S + R_2\, C_1\, C_2\, S^2 + C_1\, S}} \ \ \frac{1}{R_2\, C_2 S + 1}$$

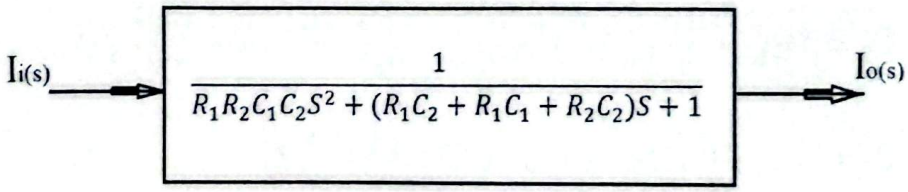
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$$\frac{V_o}{V_i} = \frac{\dfrac{R_2C_2S+1}{C_2S+R_2C_1C_2S^2+C_1S}}{\dfrac{R_1(C_2S+R_2\,C_1C_2\,S^2+1)+R_2C_2S+1}{C_2S+R_2\,C_1\,C_2S^2+C_1\,S}} \quad \frac{1}{R_2\,C_2S+1}$$

$$\frac{V_o}{V_i} = \frac{R_2C_2S+1}{R_1(C_2S+R_2\,C_1\,C_2S^2+C_1S)+R_2C_2S+1} \quad \frac{1}{R_2\,C_2S+1}$$

$$\frac{V_o}{V_i} = \frac{1}{R_1C_2S+R_1R_2\,C_1\,C_2\,S^2+R_1C_1S+R_2C_2S+1}$$

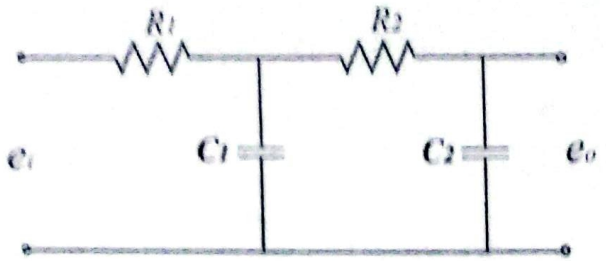
$$\frac{V_o}{V_i} = \frac{1}{R_1R_2\,C_1\,C_2\,S^2+(R_1C_2+R_1C_1+R_2C_2)S+1}$$



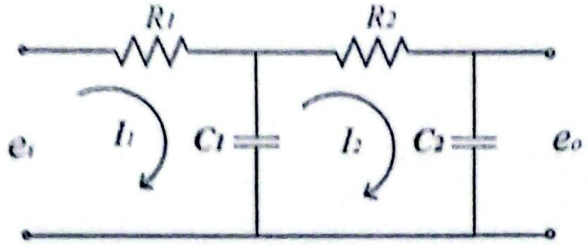
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Example (2.9) طريقة اخرى لحل السؤال

Example (2.9) : find the T.F. Transfer Function (E_o/E_i) *for the given network:*



Solution:



Using Maxwell theory of looping currents we'll have:

$$E_i = (R_1 + \frac{1}{C_1 S})\, I_1 - \frac{1}{C_1 S}\, I_2 \qquad \dots (1)$$

$$0 = -\frac{1}{C_1 S}\, I_1 + (\frac{1}{C_1 S} S + R_2 + \frac{1}{C_2 S})\, I_2 \qquad \dots (2)$$

$$E_o = \frac{1}{C_2 S}\, I_2 \qquad \dots (3)$$

From equation 2 find I_1

$$\frac{1}{C_1 S}\, I_1 = (\frac{C_2 S^2 + R\, C_1\, S\, C_2 S + C_1 S}{C_1 S\, C_2 S})\, I_2$$

$$I_1 = \frac{1}{C_1 S}(\frac{C_2 S^2 + R\, C_1\, S\, C_2 S + C_1 S}{C_1 S\, C_2 S})\, I_2$$

$$I_1 = \left(\frac{C_2 S^2 + R\, C_1\, S\, C_2 S + C_1 S}{(C_1 S)^2 C_2 S}\right) I_2 \qquad \dots (4)$$

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Substitute equation (4) in equation (1) we get

$$E_i = (R_1 + \frac{1}{C_1 S}) \left(\frac{C_2 S^2 + R\, C_1\, S\, C_2 S + C_1 S}{(C_1 S)^2 C_2 S} \right) I_2 - \frac{1}{C_1 S} I_2$$

$$E_i = \left[(R_1 + \frac{1}{C_1 S}) \left(\frac{C_2 S^2 + R\, C_1\, S\, C_2 S + C_1 S}{(C_1 S)^2 C_2 S} \right) - \frac{1}{C_1 S} \right] I_2$$

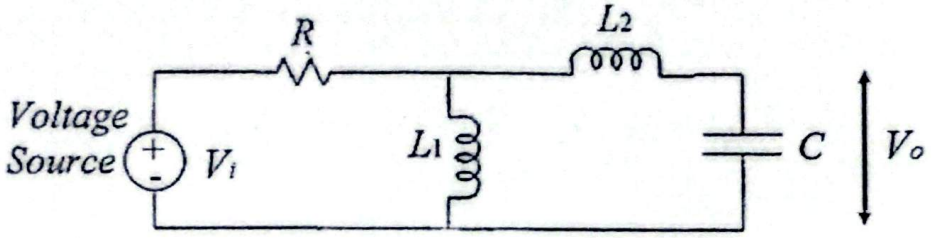
Now find Transfer Function TF = (E_o/E_i).

$$T.F. = \frac{E_o}{E_i} = \frac{\frac{1}{C_2 S}\, I_2}{\left[(R_1 + \frac{1}{C_1 S}) \left(\frac{C_2 S^2 + R\, C_1\, S\, C_2 S + C_1 S}{(C_1 S)^2 C_2 S} \right) - \frac{1}{C_1 S} \right] I_2}$$

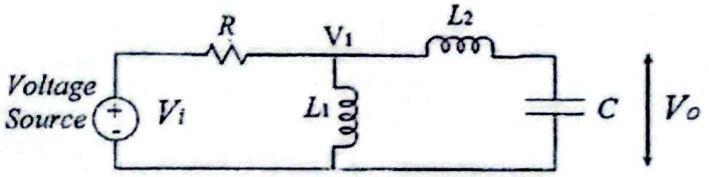
$$T.F. = \frac{E_o}{E_i} = \frac{\frac{1}{C_2 S}}{\left[(R_1 + \frac{1}{C_1 S}) \left(\frac{C_2 S^2 + R\, C_1\, S\, C_2 S + C_1 S}{(C_1 S)^2 C_2 S} \right) - \frac{1}{C_1 S} \right]}$$

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Example (2.10): *Find the T.F for the circuit shown below with the voltage source.*

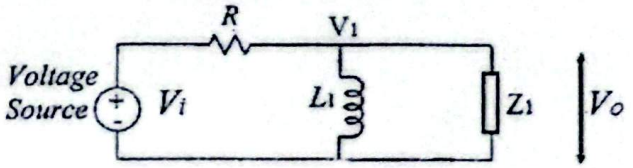


Solution:

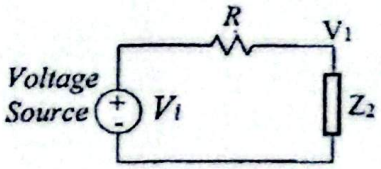


$$V_o = V_1 \frac{\frac{1}{CS}}{L_2 S + \frac{1}{CS}}$$ Voltage divider rule

$$V_o = V_1 \frac{1}{L_2 CS^2 + 1} \quad \ldots\ldots (1)$$



$$Z_1 = L_2 S + \frac{1}{CS} = \frac{L_2 CS^2 + 1}{CS}$$



$$Z_2 = L_1 S \,//\, Z_1 = \frac{L_1 S(Z_1)}{L_1 S + Z_1}$$

$$V_1 = V_i \frac{Z_2}{R + Z_2} \quad \ldots\ldots (2)$$

Substitute equation (2) in (1):

$$V_o = V_1 \frac{1}{L_2 CS^2 + 1} \quad \ldots\ldots (1)$$

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$$V_o = V_i \quad \frac{Z_2}{R + Z_2} \quad \frac{1}{L_2CS^2 + 1}$$

$$T.F. = \frac{V_o}{V_i} = \quad \frac{Z_2}{R + Z_2} \quad \frac{1}{L_2CS^2 + 1}$$

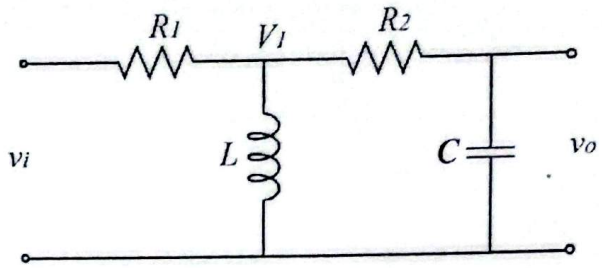
$$\frac{V_o}{V_i} = \quad \frac{\frac{L_1S(Z_1)}{L_1S + Z_1}}{R + \frac{L_1S(Z_1)}{L_1S + Z_1}} \quad \frac{1}{L_2CS^2 + 1}$$

$$\frac{V_o}{V_i} = \quad \frac{\frac{L_1S\left(\frac{L_2CS^2 + 1}{CS}\right)}{L_1S + \frac{L_2CS^2 + 1}{CS}}}{R + \frac{L_1S\left(\frac{L_2CS^2 + 1}{CS}\right)}{L_1S + \frac{L_2CS^2 + 1}{CS}}} \quad \frac{1}{L_2CS^2 + 1}$$

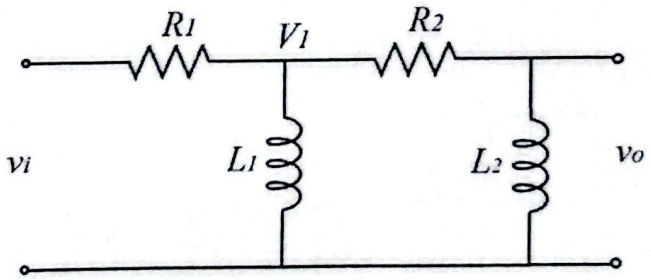
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Home Work

Q 1: *Obtain the Transfer Function* $\frac{E_o(s)}{E_i(s)}$ *of the electric circuit shown in figure below.*



Q2: *Find the T.F for the circuit shown below with the voltage source.*



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أسئلة على الفصل الثاني
Chapter Two

Q1/ Choose the correct answer from the following

1) Laplace Transforms of $y(t) = e^{-2t} + t\,e^{-5t}$ equal ----------.

a) $\frac{1}{(S-2)} + \frac{1}{(S-5)^2}$

b) $\frac{1}{(S+2)} + \frac{1}{(S+5)^2}$

c) $\frac{2}{(S+2)} - \frac{2}{(S+5)^2}$

d) $\frac{2}{(S-2)} + \frac{2}{(S-5)^2}$

Answer:b) $\frac{1}{(S+2)} + \frac{1}{(S+5)^2}$

2) The Invers Laplace Transform of $Y(S) = \frac{1}{2(S+2)} - \frac{4}{3(S-5)} + \frac{1}{(S+4)}$ equal--------.

a) $y(t) = \frac{1}{2}e^{-2t} - \frac{4}{3}e^{5t} + e^{-4t}$

b) $y(t) = \frac{1}{2}e^{2t} - \frac{4}{3}e^{-5t} + e^{4t}$

c) $y(t) = 2e^{2t} - \frac{3}{4}e^{-5t} + e^{4t}$

d) $y(t) = 2e^{-2t} - \frac{4}{3}e^{5t} + e^{-4t}$

Answer: a) $y(t) = \frac{1}{2}e^{-2t} - \frac{4}{3}e^{5t} + e^{-4t}$

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3) The Invers Laplace Transform of $Y(S) = \frac{2}{(S+2)(S+3)}$ equal ----------.

a) $y(t) = \frac{1}{2}e^{-2t} - \frac{1}{2}e^{-3t}$

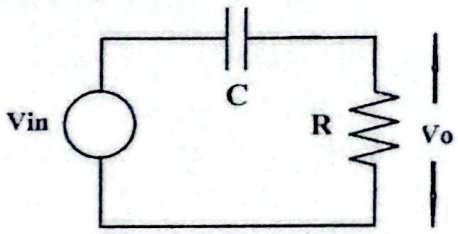
b) $y(t) = \frac{1}{2}e^{2t} - \frac{1}{2}e^{3t}$

c) $y(t) = 2e^{-2t} - 2e^{-3t}$

d) $y(t) = -2e^{-2t} + 2e^{-3t}$

Answer: c) $y(t) = 2e^{-2t} - 2e^{-3t}$

4) For the system shown below Transfer Function (V_o/V_i) equal



a) $TF = \frac{V_o}{V_i} = \frac{R}{R+CS}$

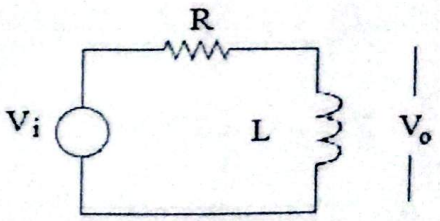
b) $TF = \frac{V_o}{V_i} = \frac{CS}{R+CS}$

c) $TF = \frac{V_o}{V_i} = \frac{R}{R+\frac{1}{CS}}$

d) $TF = \frac{V_o}{V_i} = \frac{1}{R+\frac{1}{CS}}$

Answer: $\boldsymbol{TF = \frac{V_o}{V_i} = \frac{R}{R+\frac{1}{CS}}}$

5) For the system shown below Transfer Function (V_o/V_i) equal



a) $TF = \frac{V_o}{V_i} = \frac{R}{LS+R}$

b) $TF = \frac{V_o}{V_i} = \frac{1/LS}{1/SL+R}$

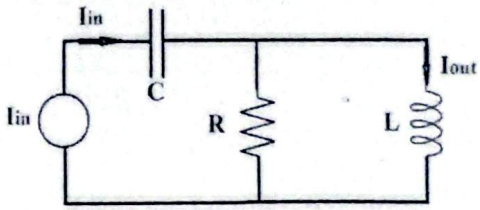
c) $TF = \frac{V_o}{V_i} = \frac{R}{\frac{1}{LS}+R}$

d) $TF = \frac{V_o}{V_i} = \frac{LS}{LS+R}$

Answer: ***d)*** $\boldsymbol{TF = \frac{V_o}{V_i} = \frac{LS}{LS+R}}$

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6) For the system shown below Transfer Function (I_o/I_i) equal



a) $TF = \frac{I_o}{I_i} = \frac{R}{LS+R}$

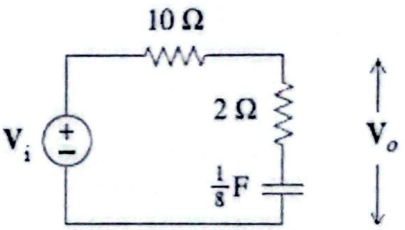
b) $TF = \frac{I_o}{I_i} = \frac{1}{LS+R}$

c) $TF = \frac{I_o}{I_i} = \frac{R}{\frac{1}{LS}+R}$

d) $TF = \frac{I_o}{I_i} = \frac{LS}{R+LS}$

Answer: a) $\boldsymbol{TF = \frac{I_o}{I_i} = \frac{R}{LS+R}}$

7) For the system shown below Transfer Function (V_o/V_i) equal



a) $TF = \frac{V_o}{V_i} = \frac{S+6}{4S+4}$

b) $TF = \frac{V_o}{V_i} = \frac{S+4}{6S+4}$

c) $TF = \frac{V_o}{V_i} = \frac{6S+4}{S+4}$

d) $TF = \frac{V_o}{V_i} = \frac{S+4}{4S+6}$

Answer: b) $\boldsymbol{TF = \frac{V_o}{V_i} = \frac{S+4}{6S+4}}$

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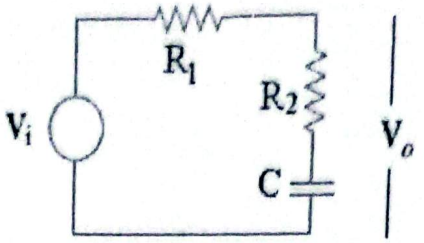
8) For the system shown below Transfer Function (V_o/V_i) equal

a) $TF = \frac{V_o}{V_i} = \frac{R_2+\frac{1}{CS}}{R_1+R_2+\frac{1}{CS}}$

b) $TF = \frac{V_o}{V_i} = \frac{R_1+\frac{1}{CS}}{R_1+R_2+\frac{1}{CS}}$

c) $TF = \frac{V_o}{V_i} = \frac{R_1+R_2}{R_1+R_2+\frac{1}{CS}}$

d) $TF = \frac{V_o}{V_i} = \frac{R_1+R_2}{R_2+\frac{1}{CS}}$



Answer: *a)* $TF = \frac{V_o}{V_i} = \frac{R_2+\frac{1}{CS}}{R_1+R_2+\frac{1}{CS}}$

9) The define of Transfer Function is

a) the ratio of the input variable to the output variable in Laplace form with all initial conduction equal zero
b) the ratio of the output variable to the input variable in Laplace form with all initial conduction equal zero
c) the ratio of the output variable to the input variable in time domain with all initial conduction equal zero
d) the ratio of the input variable to the output variable in Laplace form with all initial conduction equal one

Answer: b) the ratio of the output variable to the input variable in Laplace form with all initial conduction equal zero

10) In open loop control system, they are control systems in which

a) this system without control
b) this system is unstable
c) the output has no effect upon the control action
d) the output has an effect upon the control action

Answer: c) the output has no effect upon the control action

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11) In closed loop control system, they are control systems in which
a) the output has no effect upon the control action
b) the output has an effect upon the control action
c) this system without control
d) this system is unstable

Answer: b) the output has an effect upon the control action

12) The Objective of control system is to make the system OUTPUT and the desired REFERENCE as close as possible to make the ERROR
a) as big as possible
b) as small as possible
c) equal to one
d) greater than input reference

Answer: b) as small as possible

13) Open loop control system are
a) simple and unstable
b) simple and stable
c) accurate and reliable
d) complex and unstable

Answer: b) simple and stable

14) Closed loop control system are
a) simple and stable
b) complex and unstable
c) accurate and reliable
d) inaccurate and unreliable

Answer: c) accurate and reliable

15) SISO (Single Input and Single Output) control systems
a) have one input and one output
b) have only one input and two output
c) have one input and only two output
d) have more than one input and more than one output.

Answer: a) ***have one input and one output***

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16) MIMO (Multiple Inputs and Multiple Outputs) control systems

a) have one input and one output
b) have only one input and two output
c) have one input and only two output
d) have more than one input and more than one output

Answer: d) ***have more than one input and more than one output***

Q2/ Choose True or False for each statement

1) The define of Transfer Function is the ratio of the output variable to the input variable in Laplace form with all initial conduction equal one

True
False

2) In open loop control system, they are control systems in which the output has an effect upon the control action

True
False

3)A mathematical model of a dynamic system is defined as a set of equations that represents the dynamics of the system accurately

True
False

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